DIVYANSHU GOYAL

Multimodal Foundation Models | RL Post-Training | Agentic Evaluation

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SUMMARY

Applied Scientist & AI Tech Lead | 8+ Years of Experience

AI researcher and engineering leader with 8+ years of experience specializing in agentic AI and Multimodal LLMs. I bridge the gap between cutting-edge research and enterprise-scale products, specifically in domains where quality is dictated by nuanced human aesthetics and marketing effectiveness. During my tenure at Adobe, I led the technical strategy to scale GenStudio for Performance Marketing from 0 to 1, driving multi-million-dollar adoption across the Fortune 500. Currently, I am tech lead for a 7-person team building specialized VLMs, judge models, and retouchers. Combining product execution with rigorous research, my work includes co-leading the ATune photo retoucher (outperforming GPT-6 Astra in human preference), 6 patents, and publications at CVPR, ICML, and AAAI workshops.

EXPERIENCE

Applied Scientist – Adobe Inc., San Jose, CA June 2021 – Present

FixIt: Multimodal Models for Marketing & Visual Quality (Tech Lead, Team of 7)

- **Architectural Ownership:** Spearheaded technical vision and team execution across the **full ML lifecycle**: data curation, distributed model training, RL alignment, and low-latency inference.
- **Agentic Integration:** Developed a **sub-second** JEV-style judge integrated as the primary marketing evaluator inside the **Firefly AI Agent** harness, reducing user revision iterations by **40%**.
- **Model Portfolio:** Directed the development of **three specialized model streams**: domain-adapted VLMs, marketing-quality judges, and an unsupervised photo retoucher trained entirely without human-edited pairs.

Adobe Marketing VLM (Gemma-3-12B, SFT & GRPO Alignment)

- **Scaled RL Alignment:** Pioneered multi-round **GRPO alignment** on 8× H100s using rank-256 LoRA (1.19B trainable params), scaling policy optimization across **7 simultaneous custom reward functions** for visual quality, aesthetics, and semantic attributes.
- **Overcoming Reward Collapse:** Uncovered and resolved critical **policy collapse** where the model exploited reward ambiguity to hedge on neutral scores (84% rate vs. 56% human baseline); reshaped the reward curvature to eliminate the reward hole, restoring full discriminative range.
- **Distributed Production Stack:** Architected a fault-tolerant asynchronous annotation engine with auto-recovery and a sub-100ms router-worker inference service orchestrating real-time multimodal evaluation across 4 GPUs.

Marketing Judge Models & Benchmark Integrity

- **Calibrated Soft-Label Alignment:** Trained Qwen3.5-4B judges over a **1.1M-example dataset** using human rater disagreement distributions as soft targets, elevating KonIQ rating correlation from 0.421 to **0.663 SRCC** (paired 95% CI strictly positive).
- **Decontaminated Gating & Scale:** Built a **67K-example benchmark** verified against 537K training images via SSCD copy detection; instituted pre-registered non-inferiority gating across 10 releases and served the judge at 21 replicas on 8× H100s.
- **Vision Collapse & Selection:** Diagnosed a 200-step **vision-encoder collapse** that left checkpoints image-blind, embedding automated vision-health probes into training; reached **82% GAIC crop-selection accuracy** (+11 pts over baseline).

ATune: Unsupervised Photo Retouching (Co-Lead & Co-First Author with Adobe Research)

- **Unsupervised Control Paradigm:** Architected and co-led a **54M-parameter model** that learns parametric digital darkroom controls (RawTherapee/darktable) directly from **10M unedited photos**, completely eliminating the need for paired expert edits.
- **Outperforming Frontier LLMs:** Demonstrated state-of-the-art aesthetic edits in double-blind human evaluations (157 raters, 3,500+ trials), **beating GPT-6 Astra** (54.3% decisive win rate) and **matching Claude Opus 5** (50.5%); preferred by 8 of 9 standard aesthetic judges across FiveK, PPR10K, and SPAQ.
- **Empirical Data Scaling Laws:** Proved aesthetic control fidelity scales monotonically with uncurated data from 1M to 10M images, establishing that each larger corpus strictly dominates or ties smaller datasets across all benchmark tests.

Adobe GenStudio for Performance Marketing (Tech Lead, Team of 20+)

- **Zero-to-One Platform Architecture:** Founding tech lead for a 20+ engineer team, architecting the high-throughput core API platform for Adobe GenStudio (launched Oct 2024), scaling enterprise LLM access to generate **multi-million-dollar ARR** across **Fortune 500** brands.
- **Patented Core Intelligence:** Co-invented multimodal algorithms for structured brand-guideline extraction and real-time compliance evaluation (US20250117829), establishing the engine behind **Adobe Brand Intelligence**.
- **High-Stakes Enterprise Adaptation:** Directed custom fine-tuning pipelines on proprietary customer data for marquee Fortune 500 accounts, directly converting technical pilots into closed **enterprise custom-model contracts**.

Evaluation & Safety Research

- **DistortBench (First Author; CVPR & ICML 2026 Workshops):** Created a 13.5K-sample visual distortion benchmark across 27 categories and 5 severities, exposing critical perceptual blind spots where 18 frontier VLMs trail human majority vote.
- **JudgeBreak (ICML 2026 Workshop):** Proved that linear probing on frozen Qwen3-32B representations beats prompted LLM judges and reward models by **8.0 percentage points** (78.1% vs. 70.1%), showing evaluative taste is linearly encoded in model hidden geometry.

Machine Learning Engineer – Adobe Inc., Bangalore, India July 2016 – August 2019

- **Ultra-Low-Latency Inference Engine:** Engineered Adobe's foundational ML serving tier, achieving **3,500 QPS/node at 0.3–0.6 ms p99 latency** via a zero-GC asynchronous LMAX Disruptor architecture (Adobe Tech Blog).
- **Enterprise Lifecycle Platform:** Architected the automated model deployment platform powering versioned rollouts, real-time drift monitoring, and zero-downtime canary migrations.

PUBLICATIONS

ATune: Unsupervised Photo Retouching Through Image-Editor Controls
D. Goyal*, G. Ruiz*, A. Meisami | Under review | *equal contribution

DistortBench: Benchmarking Vision Language Models on Image Distortion Identification
D. Goyal, A. Eppa, V. Bannihatti Kumar | CVPR 2026 and ICML 2026 Workshops | arXiv:2604.19966

JudgeBreak: Is Taste Hidden in the Layers? Stress-Testing Reward Models and LLM Judges with Representation-Based Subjective Evaluation
V. Bannihatti Kumar, A. Narang, M. Anand, A. Eppa, **D. Goyal** | ICML 2026 Workshop | OpenReview

Curiosity-Driven LLM-as-a-judge for Personalized Creative Judgment
V. Bannihatti Kumar, **D. Goyal**, A. Eppa, N. Bhandari | AAAI 2026 Workshop, Oral | arXiv:2510.05135

PATENTS

Title	Number	Year
Guiding Language Translation with Translation Documents Using ML	US20260050753	2026
Brand-Aligned Marketing Content Generation (Adobe Brand Intelligence)	US20250117829	2025
Cultural Adaptation of Images Using Generative AI	US20260051084	2026
Parameter-Efficient Fine-Tuned Models	US20250217193	2025
Video-to-Video Culturalization Using Generative AI	Filed with USPTO	2026
Learning to Be Surprised: A Curiosity-Driven Framework for Creative Judgment	Filed with USPTO	2026

EXTERNAL CONTRIBUTIONS

VibeNanoChat: GPT-2 Scale LLM Pretraining from Scratch 2026 | code
Pretrained GPT-2-scale models from scratch at **>1M tokens/s** on 8× H100 using a custom distributed Muon optimizer. Derived compute-optimal scaling frontiers from isoFLOP curves over 40+ runs. Deep dives: Going Fast and Scaling Transformers.

BrownBearAI: On-Device Photo Understanding for iOS 2026 | App Store
Shipped a privacy-first iOS app executing on-device neural face clustering and deduplication at **99.4% BCubed F1** on LFW identities with zero cloud dependency.

LaSeRo: Segmenting Antarctic Rock Outcrops from Landsat 8 2019–2020 | code
Trained and evaluated SegNet architectures to segment geological rock outcrops across multispectral satellite imagery.

Face Mask Detection for COVID-19 2020 | code
Developed a real-time mask detection pipeline, benchmarking CNN architectures with VAE representation learning and Grad-CAM visual attribution.

TECHNICAL SKILLS

Research: VLM Training & Evaluation, LLM Pretraining & Scaling Laws, Benchmark Design, Human Preference Studies, LLM-as-a-Judge & Reward Models, Reinforcement Learning (GRPO), Reward Design, Linear Probing, Bootstrap Statistics

Tools: Python, PyTorch, Distributed Training (DDP, ZeRO-2), torch.compile, FlashAttention-3, Liger Kernels, Muon, Mixed Precision (BF16), LoRA, Hugging Face Transformers/PEFT, Weights & Biases

EDUCATION

Institution	Degree	GPA	Years
Georgia Institute of Technology	M.S. Computer Science (Machine Learning)	4.0/4.0	2019 – 2021
BITS Pilani	B.E. Computer Science, M.Sc. Mathematics	8.75/10	2011 – 2016